Lab 5 — Long-Run Mix and Eigen

Friday, November 20 (~25–30 min)

Last updated: November 4, 2026.

In class Friday, November 20 (same day as Lecture 30 / Quiz 10). Quiz 10 is Week 11 inverses / rank / determinants — **not** this lab.

Credit: Lab points are for effort in class, not for a Canvas file upload. Preview the one-pager; write on the hard copy handed out in class.

Today's story (discover, then explain). First iterate a campus model and notice that different starts share a long-run mix. Then use eigenvalues/vectors: that mix is the $\lambda = 1$ **eigenvector** (scaled); the other mode has $|\lambda| < 1$ and fades. Same ideas as Lecture 30.

Learning objectives

1. Iterate $x_{t+1} = Ax_t$ in software and describe a long-run pattern from two different starts.
2. Find a steady state / eigenvector for $\lambda = 1$ and scale it to a total population.
3. Match software eigenpairs to the long-run mix and explain why other modes fade.

Math connections

Lectures 28–30 (eigenvectors, characteristic equation, dominant behavior). Loop syntax from earlier labs / Lecture 22–23 dynamics.

Key syntax (this lab)

Task	Octave	Python
One step / loop	`x = A*x`	`x = A @ x`
Eigenpairs	`[V,D]=eig(A)`	`lam, V = np.linalg.eig(A)`
Diagonal entries	`diag(D)`	`lam`
Normalize / scale	`v/sum(v)`	same idea with NumPy

Part A — Discover the long-run mix

Campus on/off counts $x_t = \begin{bmatrix} c_t \\ o_t \end{bmatrix}$ with

$$A = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}.$$

C5.0. Enter A and $x_0 = (1000, 0)$. Print $x_1 = Ax_0$. Then iterate about 20 steps; record x_5 and x_{20} (or the last state). Roughly where do the populations settle?

Expected: $x_1 = (700, 300)$; toward about $(400, 600)$ by $t = 20$.

C5.1. Restart from $x_0 = (0, 1000)$. After 20 steps, what happens? In one sentence, what is surprising if you only look at the starting point?

Expected: again near $(400, 600)$. Different starts can head toward the same long-run mix.

Part B — Explain with eigenpairs

C5.2. Compute all eigenvalues of A in software. Which one has magnitude 1? Which has $|\lambda| < 1$?

Expected: $\lambda_1 = 1$ and $\lambda_2 = 0.5$ (order may differ).

C5.3. Find an eigenvector for $\lambda = 1$ (or solve $(A - I)x = 0$). Scale so the two entries sum to 1000. Compare to your x_{20} from Part A. One sentence: why does iteration settle there?

Expected: a multiple of $(2, 3)$ — e.g. $(400, 600)$. The long-run state is that $\lambda = 1$ direction; the other mode shrinks because $|\lambda_2| < 1$.

C5.4. (If time.) Capstone preview: SVD keeps the **largest** singular values. In one sentence, how is "keep the big directions" similar to "the $\lambda = 1$ mode wins the long run"?

Expected: both pick the directions that dominate the object (image vs. iterated state).

One-pager

Preview the Lab 5 ticket; write on the hard copy in class.